# The Persistence of Vision: Field-Level PII Recovery and Irreversible Redaction in Multi-Vector VLM Retrievers

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***Abstract*—An embedding is widely treated as a safe artifact: systems that would never store a scanned passport routinely store its embedding. However, multi-vector vision-language retrievers such as ColPali and ColQwen2 are now deployed over scanned forms, invoices and identity documents, keeping every page as a grid of per-patch vectors. Those vectors remain a field-level oracle for the personal data the page carried, and an erasure request that deletes the document does not reach them. An attacker with only the at-rest index and the public encoder can read them, without training or image reconstruction. A dictionary search recovers names, identity numbers and dates of birth at $0.994/0.804/0.794$ top-1 from a $200$-candidate lineup drawn from $6.8\times10^8$ real US Census and SSA names. Given no lineup at all, it still composes the correct full name on $0.147$ of documents. The leak is holographic: deleting a field's own patches, or a third of the page around them, leaves recovery at $1.00$. Worse, a learned reshaping defense reporting $1.00$ privacy against a non-adaptive attacker returns exactly the undefended rate against an adaptive one. We introduce CATARACT, an index-time orthogonal projection that destroys rather than hides the PII-discriminative subspace, certified against linear reconstruction of what it removes. It holds the strongest attacker we evaluate to $0.125$ linkage, and on a real retrieval corpus it retains $86\%$ of NDCG@5; no public corpus carries both PII labels and relevance judgements, so these are two measurements. Index-time cost is $0.64$ ms per page and no added storage. An index built from visually rich documents contains the personal data of those documents, and should be governed as such.**

***Index Terms*—multi-vector retrieval, embedding inversion, privacy, redaction**

## I. Introduction

Storing a document's embedding instead of the document is treated, almost universally, as a privacy-preserving step. The assumption underneath it is that a grid of floating-point vectors is an opaque summary rather than a copy: useful for matching, useless for reading. For the retrievers now deployed on visually rich documents that assumption is false, and the property responsible for the leak is the same one that makes them state of the art.

Visual document retrieval-augmented generation has moved decisively away from the classical pipeline of OCR, layout parsing, and text embedding. Multi-vector vision-language retrievers, ColPali on a SigLIP/PaliGemma backbone [1], ColQwen2 on Qwen2-VL [2], instead render a page as an image, encode it into a grid of per-patch embeddings, and defer all query interaction to a late-interaction $\mathrm{MaxSim}$ score inherited from ColBERT [3]. A single ColPali page becomes $1024$ image-patch vectors on a $32\times32$ grid plus six instruction-suffix tokens, each of dimension $d{=}128$, for $131{,}840$ floats stored per page. This is precisely why these systems are state of the art: retaining fine-grained per-patch evidence rather than a pooled summary captures the visual and textual detail of scanned forms, invoices, and identity documents that pooled encoders discard. That same detail is stored, at rest, in the vector index.

The question is not idle. A recent systematization of privacy in retrieval-augmented generation places *vector database leakage* in its taxonomy, then reports that beyond exploratory work on differential privacy and synthetic data it finds "relatively few works that propose mitigations at the embedding level," and closes that entry by calling for research on privacy protection in embedding representations [4]. The gap is named and left open. We ask the question the deployment pattern makes urgent: is the stored multi-vector index a personal-information oracle? The threat model is deliberately realistic (Fig. 1). The attacker has read access to the stored index, obtained through a database breach, a malicious insider, a leaked backup, or soft-deleted segments, plus the public, open-source encoder. It has neither the original page images nor any model secret; the attack is white-box only on weights anyone can download. Its goal is to link field-level PII to stored records: names, dates of birth, identity numbers.

*Scope, stated up front:* Most numbers here come from *closed-set linkage*: given a candidate lineup of size $K$, the attack identifies which candidate is present in a stored record. Recovery is therefore a function of $K$ and is read against the chance floor $1/K$; we state $K$ with every number, and we state that scope here rather than at the end because it conditions every figure that follows. Section IV-D then removes the lineup altogether: holding only public name-part vocabularies and no candidate list, the attack still composes the correct full name on $0.147$ of documents, from a space of $6.8\times10^8$. What these indices permit is therefore recovery, not only the confirmation of a name already in hand.

*The thesis: retrieval quality* is *the leak:* The result that ties this work together is an entanglement between capability and exposure. At matched storage of $128$ floats, a single ColPali patch links PII at $0.10$, *below* the pooled single-vector baseline at $0.20$. ColPali pulls ahead only as it is allowed more patches ($m{=}8 \to 0.70$, $m{=}64 \to 1.00$), and linkage rises monotoni-

cally with stored capacity across architectures with different backbones, tokenizers and patch counts (Sec. VI). The very capacity that produces state-of-the-art retrieval is the capacity that leaks. The cost of our defense (removing roughly three quarters of the embedding dimensions before privacy holds) is the direct price of that entanglement.

We make the following contributions:

1) **A field-level closed-set linkage attack** against multi-vector VLM retrievers, to our knowledge the first to link multi-vector visual-document retriever indices to field-level PII. It is training-free, requires no image reconstruction, and recovers name/ID/DOB at 0.994/0.804/0.794 top-1 from a $K$=200 lineup drawn from a real $6.8\times10^8$-name vocabulary (chance 0.005, $n$=510). A paired single-vector control (0.253/0.012/0.014; McNemar 378:0/405:1/398:0) establishes that the vulnerability is intrinsic to multi-vector encoding. Recovery holds at 0.925 against a 100,000-name lineup, and we validate on realistic identity documents from five national templates (names 0.750–0.967, wrong-page control 0.000) and on real dense forms (0.225).
2) **Open-world extraction.** The attack does not need a lineup. Given only public name vocabularies it *composes* the correct full name on 0.147 of documents from a $4\times10^7$ space (a $6\times10^6$ lift over chance) so the result is recovery, not merely membership confirmation. With no vocabulary at all it recovers nothing, which locates precisely what the attacker needs (Sec. IV-D).
3) **The holographic mechanism.** Information about each field is distributed across the whole page by full attention, so patch deletion up to a 33% dilation still recovers 1.00 and the linkage margin holds at +1.51 to +1.55 throughout. Redaction- and erasure-based privacy is therefore ineffective at the storage layer for every deletion strategy we tested, a right-to-erasure concern under GDPR Article 17 [5].
4) **Information-reshaping defenses give false privacy.** A min-max learned residual transform reaching 1.00 non-adaptive privacy is fully inverted by an adaptive attacker who knows the public defense: reconstruction cosine 0.998, and recovery returns to *exactly* the undefended baseline in every seed. Privacy must be reported attack-relative, never absolute.
5) **CATARACT**, an index-time orthogonal projection off the PII-discriminative subspace that is non-injective by construction, certified against linear reconstruction of the removed subspace and empirically robust to the strongest nonlinear inverse we train. At its knee it holds that attacker to 0.125 linkage while retaining 78% of retrieval utility on synthetic cards drawn from the real vocabulary, and retains 86.2% of NDCG@5 on a real corpus that carries no PII labels, two corpora, since none offers both measurements. It costs 0.64 ms/page, zero added storage and zero query-path overhead. We report it plainly as a mitigation: residual linkage remains roughly 25× chance.

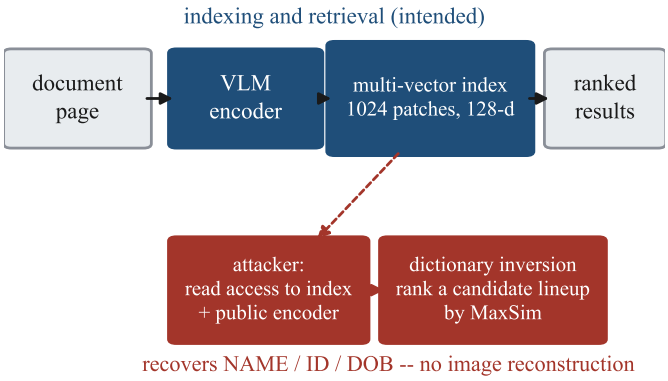



Fig. 1. Threat model. A page is encoded into a per-patch multi-vector index scored by MaxSim. An attacker with read access to that stored index and the public encoder recovers field-level PII by dictionary inversion, without reconstructing any image and without querying a live system.

The remainder of the paper is organized as follows. Section II reviews multi-vector retrieval and formalizes the threat model and privacy metric. Section III states the models, corpora and statistical protocol. Section IV develops the attack and its controls. Section V shows why pixel reconstruction is the wrong instrument for this question, and what follows for how leakage must be measured. Section VI establishes the entanglement result. Section VII characterizes the holographic mechanism. Section VIII presents the adaptive break. Section IX introduces CATARACT. Section X reports baselines, real-corpus results and deployment cost. Section XI surveys related work, and Section XII discusses scope, ethics and disclosure.

## II. BACKGROUND, THREAT MODEL, AND PRIVACY METRIC

### A. Late-interaction multi-vector retrieval

Visual document retrieval has converged on a late-interaction, multi-vector paradigm inherited from ColBERT [3] and adapted to vision-language models. Rather than compressing a page into a single embedding, the retriever encodes a page image with a VLM backbone into a grid of per-patch vectors and stores every one. A text query is likewise encoded to a sequence of token vectors, and relevance is scored by a token-to-patch maximum-similarity operator. For page patches $P_d \in \mathbb{R}^{n_p\times d}$ and a query $q \in \mathbb{R}^{n_q\times d}$,

$$S(q, P_d) \;=\; \sum_i \left( \max_j \langle q_i,\, p_j \rangle \right). \qquad (1)$$

Each query token independently selects the patch it matches best, and the per-token maxima are summed. This fine-grained matching is what makes the family state of the art on visually rich documents, where a single pooled vector cannot simultaneously represent layout, figures, and dispersed text.

We study the two dominant instances and one control. ColPali is built on a SigLIP/PaliGemma backbone and emits 1030 tokens, 1024 image patches on a 32×32 grid plus 6 instruction-suffix tokens, each of dimension $d$=128, for 131,840 stored floats per page. ColQwen2 uses a Qwen2-VL backbone and

emits roughly 267 variable-length patches per page. As a single-vector control we use BiPali, which shares ColPali's backbone but mean-pools to one 128-dimensional vector per page; it isolates the effect of multi-vector storage while holding the encoder fixed.

### B. Threat model

The attacker has *read access to the stored index*, the per-patch vectors as written to disk, and the *public encoder weights*, which for this model family are openly downloadable. It has no page images, no model secrets, and no query-time access. This corresponds to a database breach, a malicious insider, a leaked backup, or recovery of soft-deleted segments from raw storage. In the data-minimization taxonomy of Staab et al. this is ADV. A-STORE, the adversary who reaches the collector's storage container rather than the model or the query path [6].

Two properties of this threat model deserve emphasis. First, it is *at-rest*: the attack needs no interaction with a live system, so rate-limiting, query auditing and access logging do not apply. Second, it is white-box only on public weights; we show in Sec. X that an attacker using a *different* encoder fails entirely, so the assumption is load-bearing and realistic rather than convenient.

### C. Privacy metric

We measure privacy by the success of closed-set linkage. For a stored record with true field value $v^\star$ and a candidate lineup $\mathcal{V}$ with $|\mathcal{V}| = K$ containing $v^\star$, the attacker returns

$$\hat{v} \;=\; \arg\max_{v \in \mathcal{V}} \; S(\mathrm{enc}(v),\, P_d)\,, \tag{2}$$

and *recovery* is the fraction of records for which $\hat{v} = v^\star$. *Privacy* is reported as $1 -$ recovery against a stated attacker.

Three properties of this metric matter for interpreting every number in the paper. (i) It is objective: correctness is exact string equality against a known ground-truth value, requiring no human adjudication or semantic similarity judgment. (ii) It is bounded by the lineup: the chance floor is $1/K$, so recovery is only meaningful alongside $K$, which we always state. (iii) It is attack-relative: as Sec. VIII shows, the same defense can report $1.00$ privacy against one attacker and $0.00$ against another, so an unqualified privacy number is meaningless.

## III. EXPERIMENTAL SETUP

Appendix D records the provenance chain and two aggregation hazards. All experiments run inside a single container image pinned by git SHA, so every number in this paper is tied to the exact code that produced it. Each result file records the git SHA, torch and CUDA versions, and the accelerator used (NVIDIA A100-SXM4-40GB or NVIDIA L4).

*Models:* ColPali (`vidore/colpali-v1.3`; SigLIP-So400m / PaliGemma) emits 1030 tokens per page, 1024 image patches on a 32×32 grid plus six instruction-suffix tokens, at $d$=128, for 131,840 stored floats. ColQwen2 (`vidore/colqwen2-v1.0`; Qwen2-VL) emits a variable grid of roughly 267 patches. The single-vector control mean-pools ColPali's patches to one 128-dimensional vector, holding the backbone fixed so that only the stored representation differs.

TABLE I
FIELD-LEVEL LINKAGE, $n$=500 CARDS, $K$=200 LINEUP (CHANCE 0.005). COLPALI STORES 131,840 FLOATS/PAGE; BIPALI STORES 128. MCNEMAR COUNTS ARE ONE-DIRECTIONAL DISCORDANT PAIRS.

| Field | ColPali | BiPali | McNemar |
|---|---|---|---|
| name | **0.996** [0.990,1.000] | 0.180 [0.148,0.212] | 408:0 |
| ID no. | **0.834** [0.802,0.864] | 0.034 [0.018,0.052] | 400:0 |
| DOB | **0.836** [0.804,0.868] | 0.002 [0.000,0.006] | 417:0 |

*Corpora:* Three, chosen to span the range of document types the attack behaves differently on. *Synthetic identity cards* are generated with randomized fonts, sizes, positions, field order, background and noise, which removes the fixed-layout confound a reviewer would otherwise raise. *IDNet* [7] supplies realistically rendered national identity documents (genuine templates, layouts, fonts and scan artifacts) carrying synthetically generated field values; we use five national templates (ES, FI, RS, LV, SK). *FUNSD* [8] supplies real dense administrative forms.

*Metric and lineups:* Recovery is exact top-1 string match against the ground-truth value (Sec. II-C), reported with the lineup size $K$ and chance floor $1/K$. Unless stated otherwise $K$=200, chance $0.005$. On IDNet, distractors are drawn from the corpus's own value distribution so the lineup is in-distribution by construction.

*Statistics:* Point estimates carry $95\%$ percentile-bootstrap confidence intervals over 2000 resamples. Headline results are repeated over three seeds and reported as across-seed mean with a bootstrap CI, with Holm–Bonferroni correction across the metric family. Where a result is single-seed we say so.

*Hardware:* Decisive experiments ran on NVIDIA A100-40GB; follow-up runs used NVIDIA L4. Both support bfloat16 in hardware, which the pinned image uses. Index-time defense cost is measured on CPU, since that is where it would be applied in deployment.

## IV. THE DICTIONARY-INVERSION ATTACK

Sec. III gives models, corpora and statistics. The attack itself is deliberately minimal. It trains nothing, reconstructs nothing, and uses only the public encoder and the stored patch vectors. For each candidate value it encodes the string, scores it against the stored page by Eq. 1, and returns the top-ranked candidate (Eq. 2). That such a trivial procedure succeeds is itself the finding: no inversion machinery is required, because the stored representation already discriminates field values directly. We arrived at it only after three generative reconstruction attacks failed and produced a confounded null; Sec. V records that dead end, because a generative failure cannot establish the absence of information.

### A. Multi-vector encoding is the vulnerability

Table I reports the head-to-head against the pooled control on 500 synthetic identity cards, all three encoders sharing a backbone and differing only in what is stored.

The McNemar discordance is essentially one-directional: on the synthetic vocabulary the pooled control recovers a value the multi-vector index misses in 0 of 1,225 discordant pairs, and on the real vocabulary in 1 of 1,181. We state the exception rather than the absolute because the absolute is what a reader would check. The asymmetry is a qualitative architectural difference, not a capacity gradient along a shared curve, and it is the reason we frame the paper around multi-vector storage rather than around vision-language encoders in general.

### B. Controls: the attack is genuine

Three controls rule out the obvious artifacts; Appendix C gives all of them in full, including one we discarded.

*Wrong-page:* Scoring a record's true value against a *different* document's patches must collapse to chance, or the attack is matching template structure rather than content. It yields exactly $0.000$ ($n$=50, chance $0.005$).

*Anagram:* If $\mathrm{MaxSim}$ were bag-of-tokens matching, a character permutation of the true value would score identically. The true value beats its own permutations at $1.00$ for names and $0.93$–$1.00$ for dates, confirming that contextual query tokens preserve order. Identity numbers recover at $0.77$–$0.87$, some $160\times$ chance, and the residual errors are diagnostic rather than random: they are digit transpositions, the right multiset in the wrong order, so even a miss narrows the field to a handful of permutations.

*Lineup scaling:* Recovery must degrade gracefully as the candidate space grows, and it does: $1.000 \to 1.000 \to 1.000 \to 0.925$ at $K = 10^2, 10^3, 10^4, 10^5$, a lift of $92{,}500\times$ over chance at the largest lineup. Every candidate at every $K$ is a real GIVEN SURNAME pair drawn from US Census surnames and SSA given names, so the $K$=$10^5$ lineup is a genuine 100,000-way discrimination among real identities rather than a small vocabulary padded out to look large.

That last point is a correction, and we state how it was found. This sweep previously drew distractors from a closed 240-name vocabulary and, once exhausted, extended it with procedurally generated pseudo-names. Two artifacts followed. Early versions padded with middle initials, and because $\mathrm{MaxSim}$ *sums* over query tokens a longer candidate outscored the true value on token count alone. Fixing that left a subtler one: the generated names were title-cased while every true value was upper-cased, so at $K$=$10^5$ only 240 of 100,000 candidates were upper-case and an attacker could discard $99.76\%$ of the lineup on capitalization alone, without reading a stored vector. The effective chance floor was $1/240$, not $1/10^5$. We had already discarded one control for exactly this reason (Appendix C) and had not applied the same scrutiny here.

We re-ran both protocols to measure what the shortcut was worth rather than argue about it. On the flawed pool recovery is $1.000/1.000/0.975/0.950$; on the homogeneous real pool it is $1.000/1.000/1.000/0.925$. At $n$=40 a single document is $0.025$, so the two differ by exactly one document at the two largest lineups, in opposite directions. The shortcut was available to an attacker but ours was not taking it: $\mathrm{MaxSim}$ scores against the stored patches, and capitalization was never what it was reading.

Every candidate in every lineup we report is therefore drawn from one homogeneous pool, and the released sweep prints a homogeneity check on each run so the property can be verified rather than assumed.

TABLE II

FIELD-LEVEL LINKAGE ON THE REAL VOCABULARY ($n$=510 OVER THREE SEEDS, $K$=200, CHANCE 0.005). NAME RECOVERY IS INDISTINGUISHABLE ACROSS VOCABULARIES; IDENTITY NUMBERS AND DATES FALL BY THREE TO FOUR POINTS ON THE REAL ONE. BIPALI IS THE PAIRED SINGLE-VECTOR CONTROL AT THE SAME LINEUPS; MCNEMAR COUNTS ARE DISCORDANT PAIRS (COLPALI-ONLY : BIPALI-ONLY) POOLED OVER THE THREE SEEDS. THE LAST COLUMN IS THE SYNTHETIC-VOCABULARY RESULT OF TABLE I.

| Field | ColPali | 95% CI | BiPali | McNemar | synth. |
|---|---|---|---|---|---|
| name | **0.994** | [0.982,1.000] | 0.253 | 378:0 | 0.996 |
| ID no. | **0.804** | [0.794,0.818] | 0.012 | 405:1 | 0.834 |
| DOB | **0.794** | [0.759,0.818] | 0.014 | 398:0 | 0.836 |

### C. A real name vocabulary

The synthetic cards above draw names from a hand-built vocabulary of 16 given names and 15 surnames. That is 240 names, so a $K$=200 lineup is $83\%$ of the entire space the victims are drawn from, and a reader is entitled to ask whether the result is a property of multi-vector storage or of a small, phonetically distinctive word list.

We repeat the measurement on a real vocabulary: 100,000 surnames from the 2010 US Census and 6,782 given names from SSA records, both public domain and both shipped with the artifact. The combined space is $6.8\times10^8$ names, frequency-ordered, so the head of the list is dense in exactly the confusable cases a hand-built list omits, common surnames sharing prefixes, given names differing by one character.

*The attack barely depends on the vocabulary:* Table II reports the result, and name recovery is statistically indistinguishable between the two settings ($0.996$ against a real-vocabulary interval of $[0.982, 1.000]$). Identity numbers and dates fall by three to four points ($0.834 \to 0.804$, $0.836 \to 0.794$), the direction a denser vocabulary of confusable values predicts, and we adopt the lower real-vocabulary figures as the headline. The lineup-scaling sweep makes the same point at scale, since its $K$=$10^5$ pool is drawn entirely from this vocabulary (Sec. IV). Whatever the attack is reading, it is not an artifact of a small word list.

*The defense does:* The same substitution moves the defense, and in the costly direction: at $k$=96, CATARACT holds the adaptive attacker to $0.147$ residual linkage at $0.761$ utility retention with the same 180-name fitting set used on the synthetic vocabulary, against $0.108$ at $0.875$ there. Both figures are at $n$=200 victims unless stated; the synthetic comparison is at $n$=40, the largest we ran before the real vocabulary made a larger victim pool possible.

The asymmetry is the informative part. The attacker scores a lineup it is handed; the defender must *estimate* the subspace to remove, from a sample of fitting names, and then generalise

to victims sharing no token with that sample. On the synthetic vocabulary those 180 fitting names covered 75% of the space the victims came from; on the real one they are a vanishing sample of it, and the victims share no token with it at all. Enlarging the fitting set improves both sides of the trade at once: at $n$=200 victims, fitting on 1,000 names rather than 180 moves residual linkage from 0.147 to 0.125 *and* utility retention from 0.761 to 0.785. Better estimation of the subspace is therefore free of the usual privacy–utility tension, and an operator with more unlabelled pages gets both.

We therefore headline the real-vocabulary operating point, because it is the one a deployment would actually obtain. Every defense number in this paper is quoted at that harder setting rather than at the more favourable synthetic one.

### D. Open-world extraction: what the lineup was hiding

Every number so far is closed-set, and the obvious objection is that closed-set linkage is membership confirmation rather than recovery. We take it seriously enough to remove the lineup.

The attacker gets no candidate list. It holds only public name-part vocabularies (US Census surnames and SSA given names, both freely downloadable) scores each part independently against the stored patches, and recomposes the best candidates. The answer must be *built*, from a space of $2{,}000 \times 20{,}000 = 4{\times}10^7$, rather than selected from $K$ options.

It recovers the exact full name on 0.147 of documents (95% CI $[0.120, 0.200]$, three seeds) against a chance floor of $2.5{\times}10^{-8}$, a lift of order $10^6$. The stored patches do not merely confirm a name presented to them; they emit one.

*This is a floor, not a ceiling:* One property of our search, not of the leak, costs it documents outright: given names and surnames are scored independently and the two public vocabularies overlap, so a strongly-scoring token can win both slots. Four percent of predictions are degenerate repeats (CLYDE BARAKAT → CLYDE CLYDE), and every one is a guaranteed miss.

Search width, against expectation, is *not* the constraint. Widening the beam from 24 to 64 (4,096 recompositions per document instead of 576) leaves exact recovery unchanged at 0.200. The limit is structural rather than computational: independent part-scoring cannot recover a name whose halves do not each rank in the top $B$, and a wider $B$ does not change which parts rank. A *joint* scorer, not a larger one, is the direction that would raise this number. We report the measured rate rather than an optimized one because the claim is that the capability exists, not that we have found its limit.

Exact match also understates what is recovered. The near misses are orthographic neighbours of the truth, GOBEL/GOEBEL, REILLY/REILLEY, FREDRIC/FREDERICK, which is what a method reading rendered glyphs produces and what a spurious one does not. Counting predictions within two edits raises recovery to 0.280 on the seed we inspect.

*What the attacker actually needs:* Removing the vocabulary as well collapses the attack completely. Beam search over the raw character alphabet, with no word list of any kind, recovers nothing (0.000 exact; character accuracy 0.049 against a $1/27$ floor). The four regimes bound the capability:

| attacker's prior | recovery |
|---|---|
| lineup of $K$=200 | 0.996 |
| lineup of $K{=}10^5$ real names | 0.925 |
| public name vocabularies, no lineup | 0.147 |
| none (raw alphabet) | 0.000 |

What the attack needs is therefore a public word list, and nothing beyond it. A stored patch grid does not speak unprompted; a freely downloadable Census file is the whole of the prompt, and that is not a barrier any operator can rely on. Measuring all four regimes locates the boundary precisely rather than leaving it to be assumed in either direction.

*A note on the length assumption:* Character decoding runs at the target's true length. MaxSim *sums* over query tokens, so a longer query scores higher for length alone; comparing hypotheses across lengths would measure length rather than leakage, the same artifact that made middle-initial padding beat real distractors in an early version of our lineup sweep (Sec. IV). Fixing the length makes each beam step a like-for-like comparison. The attacker is given the target's length and nothing about its content, a strictly weaker assumption than a lineup, and we state it as one.

### E. Realistic identity documents

Synthetic cards alone cannot establish external validity, so we evaluate on IDNet [7], whose documents are realistically *rendered* (genuine national templates, layouts, fonts and scan artifacts) while being synthetically *populated*. This tests the attack against realistic imagery without processing anyone's actual personal data. Distractors are drawn from IDNet's own value distribution, so the lineup is in-distribution by construction.

TABLE III
LINKAGE ON REALISTIC IDENTITY DOCUMENTS ($n$=60 PER LOCATION, $K$=200, CHANCE 0.005). NAMES TRANSFER ACROSS NATIONAL TEMPLATES; IDENTITY NUMBERS DEPEND STRONGLY ON THE RENDERED FORMAT.

| Location | name | ID no. | DOB |
|---|---|---|---|
| Spain (ESP) | 0.750–0.917 | 0.900 | **0.867** |
| Finland (FIN) | 0.950 | 0.217 | 0.700 |
| Serbia (SRB) | 0.933 | 0.283 | 0.683 |
| Latvia (LVA) | 0.850 | 0.817 | 0.400 |
| Slovakia (SVK) | **0.967** | **0.933** | 0.767 |
| *wrong-page control* | *0.000* | *0.017* | *0.000* |

Two observations matter. First, **name recovery is stable at** 0.750**–**0.967 **across five unrelated national templates**, so the attack is not an artifact of one document design. Second, **identity-number recovery spans more than fourfold** across countries, from 0.933 and 0.900 for Slovakia and Spain to 0.217 and 0.283 for Finland and Serbia, with Latvia at 0.817. This is consistent with the glyph-height dependence characterized below: the attack's strength is a property of how a field is

*rendered*, not of which authority issued it. We therefore report identity-number recovery on real documents as a range rather than a point.

Crucially, the wrong-page control also holds on real imagery. Every ESP document shares a single national template, so high recovery could in principle reflect layout; scoring true values against other documents' patches collapses to 0.000/0.017/0.000. The realistic-document results are controlled exactly as tightly as the synthetic ones.

### F. What bounds recovery: the glyph-height law

On FUNSD [8], a corpus of dense, small-font administrative forms, recovery falls to 0.225 top-1 (0.673 top-5, $K$=20, chance 0.05). Splitting by field length explains why: long fields reach 0.303 while fields of eight characters or fewer sit at 0.056, essentially chance. Recovery scales with glyph height, and the two real corpora pin both ends of that curve: FUNSD is its low end, the identity-document regime of Table III its high end. The governing variable is therefore *how large a field is rendered*, not whether the document is real, and an operator can read off the exposure of a document class from the size at which its fields are printed. Identity documents, which print exactly the fields that matter at exactly the sizes that leak, sit at the exposed end by design.

## V. Why the Obvious Attack Is the Wrong One

We report this in the body rather than in an appendix because it establishes how leakage has to be measured in this setting, not merely how we happened to proceed. A reconstruction that returns nothing is routinely read as evidence that an embedding holds nothing. That reading is unsafe, and we show why.

The intuitive way to ask whether an embedding retains personal data is to try to invert it back into the page. We spent three attacks doing exactly that. A convolutional decoder trained on real form pages reduced its loss by 2.76× and produced blurry structure with no legible text. A latent-diffusion variant collapsed to blank output: scanned documents are roughly 90% white, so the reconstruction loss was minimised by emitting nothing, and the discriminator saturated. Adding an ink-weighted loss and a gradient penalty fixed the collapse and placed ink at the correct locations, but never formed glyphs, the autoencoder downsamples 8×, and a 1–2 px stroke cannot survive an 8×8 px latent cell.

We had pre-registered a comparison to decide whether to continue: reconstruct from the multi-vector retriever and from the pooled single-vector control, and compare fidelity. The result was a *stop* decision. Both sat at the same floor, 0.008 by our reconstruction metric, indistinguishable from each other.

Read at face value, that is evidence that multi-vector storage leaks no more than pooled storage, which is the opposite of this paper's finding.

It was a confounded null. The instrument could not express the information regardless of whether the information was present, so it returned the floor for both conditions. Replacing generative reconstruction with a discriminative probe on the same stored vectors showed near-perfect field discrimination immediately, and the dictionary attack of Sec. IV followed directly. Nothing about the models or the data changed; only the instrument did.

TABLE IV
LINKAGE AGAINST MEASURED STORED CAPACITY ($n$=120, $K$=200). UNSATURATED FIELDS RISE MONOTONICALLY WITH RETAINED CAPACITY.

| Model | floats/page | name | ID no. | DOB |
|---|---|---|---|---|
| pooled control | 128 | 0.125 | 0.025 | 0.008 |
| ColQwen2 | 34,176 | 1.000 | 0.717 | 0.642 |
| ColPali | 131,840 | 0.992 | **0.833** | **0.842** |

The claim that generalizes is a sharp one: **a failed reconstruction cannot establish the absence of information.** It bounds only what that reconstruction could express. Privacy assessments that certify an embedding as safe on the strength of poor reconstruction quality (a common pattern) are measuring the decoder, not the representation. A discriminative or retrieval-based probe is the stronger instrument, and in our case it moved the measured leak from "at the floor" to 0.996.

## VI. Entanglement: Retrieval Quality *is* the Leak

A natural reading of Table I is that late interaction is peculiarly dangerous. A matched-storage control shows otherwise, and the correction matters because it constrains every defense that follows.

Restricting ColPali to a single retained patch (128 floats, exactly BiPali's budget) drops linkage to 0.10, *below* the 0.20 the pooled vector reaches on that same single-patch comparison (the pooled control of Table IV, measured on a different corpus at $n$=120, links names at 0.125). Per stored byte, mean-pooling is the more revealing representation. ColPali only overtakes it once permitted more patches: $m$=8 → 0.70, $m$=64 → 1.00. The advantage is capacity, not late-interaction magic.

Fig. 2 and Table IV extend this across the family, confirming it is a property of the class rather than of one model. Holding the documents and lineups fixed so no model benefits from an easier draw:

Identity numbers and dates rise cleanly with capacity (0.717 → 0.833 and 0.642 → 0.842) across backbones with different tokenizers and patch counts. Name recovery is *saturated* at both multi-vector capacities and therefore carries no signal here; we judge monotonicity only on unsaturated fields and say so explicitly rather than reporting a single aggregate.

The consequence is the paper's organizing constraint. Because exposure tracks retained capacity, and retained capacity is exactly what delivers state-of-the-art retrieval, no defense can remove the leak while preserving the property that motivates the architecture. It can only trade along the frontier. The cost of our defense (removing roughly three quarters of the embedding dimensions) is the price of that entanglement, not an artifact of our construction.

## VII. The Holographic Mechanism

Where does the recoverable information physically reside? The answer determines whether any localized remedy, masking

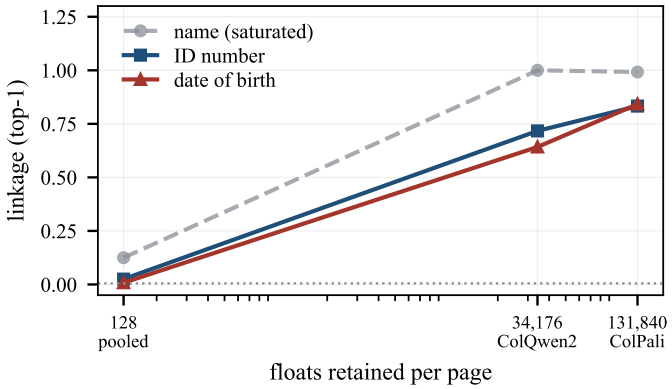



Fig. 2. Linkage against measured stored capacity, identical documents and lineups across models. Unsaturated fields rise monotonically with retained capacity; *name* is at the ceiling on both multi-vector models and therefore carries no signal here.

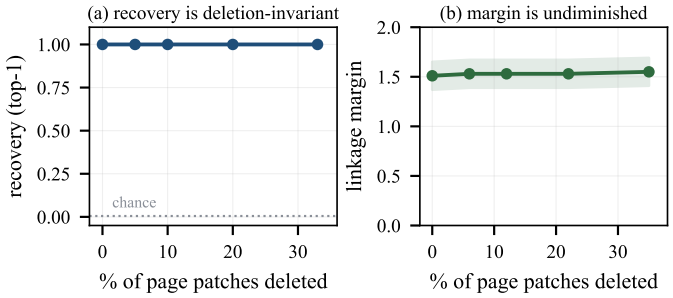



Fig. 3. The leak is holographic. (a) Recovery stays at 1.00 as up to a third of the page's patches are deleted, including the field's own. (b) The linkage margin is undiminished across the same sweep, so the correct identity is not merely winning but winning comfortably; recovery alone could not distinguish those cases.

a field's pixels, deleting the patches that overlie it, or honoring an erasure request, can work.

The mechanism is context bleed. A vision-language encoder processes all 1024 patches through stacked self-attention, so the final vector at any grid position is a function of the whole page rather than its local receptive field. Each field's identity therefore leaves a trace in patches far from where its glyphs are rendered.

Fig. 3 reports two ablations. Deleting every patch whose grid cell overlaps the rendered name leaves recovery *unchanged* at 1.00: the vectors that most obviously "contain" the name are unnecessary to recover it. Dilating the deletion to the field plus a radius-8 neighborhood removes 341 of 1024 patches (33% of the page) and recovery still holds at 1.00.

This is not a top-1 result balanced on a knife edge. The linkage margin, the true value's score minus the best distractor's, holds at $+1.51$ to $+1.55$ across the entire deletion sweep. The correct identity is not merely winning; it is winning comfortably after a third of the page has been destroyed. Only when deletion approaches the whole page does recovery finally decay.

The property is not idiosyncratic to one backbone. ColQwen2, with a different backbone, tokenizer and a variable grid of roughly 267 patches, reproduces the signature: leak 0.975, and randomly removing 90% of patches still recovers 0.83. Nine tenths of the representation can be discarded and the name survives.

We are careful about what recovery alone can show. Because top-1 is pinned at 1.00 across the sweep, it cannot by itself distinguish a truly distributed code from one that degrades gracefully; the stable margin is what establishes that the leak is robust rather than marginal. We therefore describe these results as consistent with, but not a proof of, a fully distributed representation.

*Consequence for erasure:* A GDPR Article 17 [5] erasure implemented as local redaction (blacking out the field, or dropping its patches at index time) leaves the erased quantity present, at full fidelity, in the patches left behind. Every localized remedy we tested failed. This is an information-placement finding, not a tuning failure.

TABLE V
ADAPTIVE ATTACKS ON THE RESHAPING DEFENSE. ONLY INVERSE LEARNING IS NEEDED; RECOVERY RETURNS TO THE UNDEFENDED RATE.

| Strategy | Recovery | Breaks defense? |
|---|---|---|
| Distillation probe | 0.000 | no |
| **Inverse learning** | **0.975** | **yes** |
| Adaptive query | 0.000–0.025 | no |
| Correlation | 0.000 | no |

## VIII. False Privacy: Reshaping Defenses Invert

Section VII leaves a narrow design space: if information cannot be removed locally, a defense must act globally on every stored patch. The natural construction is a learned transform that reshapes the embedding to suppress PII while preserving content. We built exactly that, and it fails in an instructive way.

Our RedactionProjection is a per-patch residual transform $y = x + \gamma\,\mathrm{MLP}(x)$, applied at index time and trained min-max: a utility term preserves topic ranking while a privacy term pushes true-value MaxSim below distractors. Against the non-adaptive dictionary attack it is a complete success, privacy 1.00 at utility 0.95, dominating flat Gaussian noise by $+0.74$ at matched privacy.

That number is an illusion. Under Kerckhoffs's assumption the attacker knows the public defense and can train an inverse. Table V evaluates four adaptive strategies against it:

The inverse learner reconstructs the original patches at cosine 0.998 and recovery returns to 0.975 (95% CI $[0.925, 1.000]$, $n$=3 seeds). The sharpest statement of the result is this: **in every seed the adaptive recovery equals the undefended baseline exactly** ($1.000/1.000$, $1.000/1.000$, $0.925/0.925$). The defense does not partially leak. It returns the index to *fully undefended* recovery while reporting 1.000 privacy (with zero variance) against the weaker attacker.

Two conclusions follow, and they shape the rest of the paper. First, privacy for this class must be reported attack-relative; an unqualified privacy number is meaningless, since the same defense scores 1.000 and 0.025 depending only on who is asking. Second, and constructively: a transform that

*reshapes* information leaves it present and therefore recoverable. A defense must destroy information, not hide it.

## IX. CATARACT: DESTROYING THE PII SUBSPACE

Section VIII leaves one viable design. A defense cannot act locally, because the leak is holographic; and it cannot reshape, because reshaping is invertible. It must be *non-injective*: it must destroy the information it is asked to protect, so that the protected component is absent from the stored vector rather than concealed within it. The distinction is exact rather than rhetorical. A transform that is many-to-one on a subspace maps every input differing only within that subspace to the same output, so no function of the output, linear or otherwise, can return the component that differed. An adaptive adversary may still *predict* that component from the part we retain, which is a statement about correlation in the data and not about inverting the transform; we separate the two carefully below.

CATARACT is the minimal construction with that property. We estimate the low-dimensional subspace along which name-value queries discriminate between records, and project every stored patch orthogonally off it. The name is metaphorically apt: the index is clouded so that it can no longer resolve identity, while retaining enough structure to keep finding documents.

### A. Construction

Let $N$ be the matrix of encoded name-value queries and $V_r$ the top-$r$ directions spanned by topic queries, which carry legitimate retrieval signal we must spare. We remove the topic component from the name signal and take the leading $k$ directions of the residual:

$$D \;=\; \mathrm{top}_k \,\mathrm{SVD}\big(N - NV_rV_r^\top\big)\,, \tag{3}$$

and redact each stored patch by orthogonal projection onto $\mathrm{span}(D)^\perp$, renormalized:

$$x' \;=\; \frac{(I - DD^\top)\,x}{\|(I - DD^\top)\,x\|}. \tag{4}$$

The transform is applied once, at index time, to the bytes actually written. It adds no parameters to the query path and no bytes to storage.

### B. What is, and is not, certified

Eq. 4 is a rank-deficient orthogonal projection, so the removed component is not merely hidden but absent. For any linear reconstruction $g$,

$$\forall g: \quad \mathbb{E}\,\|x_\parallel - g(\Pi x)\|^2 \;\geq\; \|DD^\top x\|^2, \tag{5}$$

where $\Pi$ is the projection of Eq. 4 and $D$ is as defined in Eq. 3; empirically the optimal linear inverse recovers $0\%$ of $\mathrm{span}(D)$ (span error $1.000$ for all $k > 0$).

The removal itself is irreversible in the strongest sense available: $\Pi$ maps $x$ and $x + v$ to the same point for every $v \in \mathrm{span}(D)$, so no function of the stored vector (linear, nonlinear, or otherwise) can distinguish them. What Eq. 5 adds is the quantitative floor for linear $g$.

TABLE VI
CATARACT $k$-SWEEP UNDER THE ADAPTIVE INVERSE ATTACKER ($d$=128, $K$=200, CHANCE 0.005), ON SYNTHETIC CARDS AT ONE SEED; THE $k$=96 ROW IS CONFIRMED ACROSS THREE SEEDS IN TABLE VII. THE RESHAPING DEFENSE OF SEC. VIII IS SHOWN FOR REFERENCE. AT $k$=128 EVERY DIRECTION IS REMOVED, SO UTILITY IS 0 AND ADAPTIVE PRIVACY IS UNDEFINED.

| $k$ removed | Non-adaptive priv. | **Adaptive priv.** | Utility |
|---|---|---|---|
| reshaping (ref.) | 0.950 | **0.025** | 0.950 |
| 32 | 0.700 | 0.600 | 0.925 |
| 64 | 0.900 | 0.675 | 0.900 |
| 80 | 0.900 | 0.700 | 0.875 |
| **96 (knee)** | 0.925 | **0.900** | **0.875** |
| 112 | 0.975 | 0.875 | 0.650 |
| 128 | 1.000 | — | 0.000 |

A trained *nonlinear* model nonetheless reproduces roughly $0.73$ of the removed span component, and it is worth being exact about what that is. It is not an inverse: the removed component is absent, and nothing can invert a many-to-one map. The model is *predicting* the typical removed component from the *retained* one, using the correlation between them in the data. That it succeeds is not a gap in the guarantee, it is the entanglement of Sec. VI appearing again, this time as the statement that what CATARACT removes was partly redundant with what it keeps. End-to-end linkage still collapses, because reconstructing a direction is not recovering the discrete value that direction encoded. CATARACT is therefore **provably robust to linear inversion and empirically robust to the strongest nonlinear inverse we train**, and that is precisely the claim we make.

### C. The privacy dial

Fig. 4 and Table VI sweep the removed rank $k$ against the same adaptive inverse learner that broke the reshaping defense.

Appendix B gives the full sweep and the per-field subspaces of Table VIII. At $k$=96 the attacker that drove the reshaping defense to $0.025$ privacy is held to $0.892$ across three seeds (95% CI $[0.875, 0.900]$) at utility $0.875$ (CI $[0.875, 0.875]$), with the corrected verdict independently selecting $k$=96 in every seed. On the real vocabulary the same operating point costs more to reach, and how much depends on how well the subspace is estimated: $0.147$ residual linkage from $180$ fitting names against $0.125$ from $1{,}000$, at $n$=200 victims. Several numbers are therefore attached to $k$=96, on different corpora and different metrics. Table VII states them in one place, so that none is mistaken for a competing value of the same quantity.

*A better estimator is not the lever:* If $k{\approx}96$ of 128 dimensions seems a large budget, the natural suspicion is that the estimator is spending it badly. (3) takes the directions the name queries *vary* along, and much of that variance is shared by every name, it encodes "this is a name-shaped query" rather than which name, so a discriminative criterion should need fewer directions. We built that estimator, maximizing between-identity over within-identity scatter, and it is decisively worse: on the real vocabulary at $180$ fitting names, $k$=96 holds the

TABLE VII

THE $k$=96 OPERATING POINT, STATED ONCE. ROWS DIFFER BY CORPUS AND BY METRIC, NOT BY DISAGREEING MEASUREMENTS OF ONE QUANTITY. BRACKETED INTERVALS ARE 95% PERCENTILE-BOOTSTRAP CIS OVER THREE SEEDS. PRIVACY AND UTILITY ARE ALWAYS MEASURED ON DIFFERENT CORPORA, BECAUSE NO CORPUS WE KNOW OF CARRIES BOTH PII LABELS AND RELEVANCE JUDGEMENTS (SEC. XII).

| Quantity | Corpus | Value |
|---|---|---|
| Adaptive linkage | synth. cards, 3 seeds | 0.108 [0.100, 0.125] |
| Adaptive privacy | synth. cards, 3 seeds | 0.892 [0.875, 0.900] |
| Adaptive linkage | real vocab., $n$=200 | 0.125 |
| Utility, topic R@1 | synth. cards, 3 seeds | 0.875 [0.875, 0.875] |
| Utility, NDCG@5 ret. | ViDoRe [1] | 86.2% |
| Utility, R@1 ret. | synth., 200 docs | 78% |
| Index-time cost | CPU, per page | 0.64 ms, +0 B |

adaptive attacker to 0.192 privacy against the SVD estimator's 0.833 in that same setting, and at $k \leq 64$ to essentially none. The reason is open-set generalization. Fisher directions separate the identities they were fit on; the victims share no token with that fitting set, and separations tuned to 180 specific people do not transfer. Variance directions describe the name manifold itself and do transfer. **The budget is not being wasted, it is the price of removing a subspace that generalizes.**

*Nor is sparing more content:* CATARACT spares the top $r$ topic directions before choosing what to remove, and every result above uses $r$=8. Sparing more should shrink the overlap with the content signal retrieval needs, so we swept it ($n$=200 victims, three seeds). It does exactly that, and pays for it in exactly the wrong currency. At $k$=96, raising $r$ from 8 to 32 lifts utility retention from 0.785 to 0.992 (content almost fully preserved) while residual linkage rises from 0.125 to 0.863, nearly the undefended rate. The two move together because they are the same directions: protecting what retrieval uses protects the identity with it. No setting of $r$ improves on $r$=8 at any $k$ we test. This is the entanglement of Sec. VI observed inside the defense's own geometry, and it is why the cost of CATARACT is not an engineering shortfall that a better estimator or a gentler projection could remove.

The cost is the thesis restated. Three quarters of the embedding dimensions must be removed before privacy holds. Low-rank removal barely dents the leak ($k \leq 32$ gives 0.600), because the PII is holographically distributed rather than confined to a few directions. Utility holds until $k \approx 96$ and then cliffs. That $k$ is not a hyperparameter we tuned for convenience; it is the price of the entanglement established in Sec. VI.

### D. The PII subspace is shared, not per-field

$D$ is estimated from name-value queries alone, which raises an obvious question: does it protect only names? Table VIII fits a separate subspace per field and cross-evaluates.

The off-diagonal entries carry the result. A subspace fit on names alone drops identity numbers from 0.775 to 0.025 and dates from 0.925 to 0.125, without ever seeing either field. This is independent evidence for the holographic mechanism of

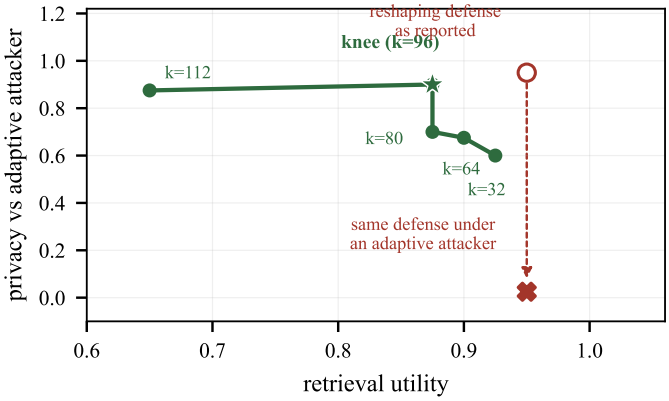



Fig. 4. The privacy–utility frontier under an *adaptive* attacker. The reshaping defense appears at the top right as reported, and collapses to the bottom right once the attacker is adaptive. CATARACT traces a tunable frontier with a knee at $k$=96.

TABLE VIII

PER-FIELD CATARACT SUBSPACES ($k$=96 EACH) ON SYNTHETIC CARDS, EACH EVALUATED AGAINST ALL THREE FIELDS. SUPPRESSING ONE FIELD SUBSTANTIALLY SUPPRESSES THE OTHERS, SO THE DISCRIMINATIVE DIRECTIONS ARE LARGELY SHARED.

| Subspace fit on | name | ID no. | DOB | Utility |
|---|---|---|---|---|
| none (undefended) | 0.975 | 0.775 | 0.925 | 0.925 |
| name | **0.025** | 0.025 | 0.125 | 0.850 |
| ID no. | 0.100 | **0.000** | 0.050 | 0.800 |
| DOB | 0.300 | 0.025 | **0.025** | 0.775 |
| combined (rank 123) | 0.000 | 0.000 | 0.000 | 0.325 |

Sec. VII: if each field occupied its own directions, a name-fitted projection would leave the others untouched. It also explains the real-document result in Table IX, where a name-fitted $D$ suppresses all three fields to near chance.

The combined subspace drives all three fields to 0.000 recovery, but consumes rank 123 of 128 and drops utility to 0.325. We say "to zero recovery" rather than "removes completely": what is measured is that the attacker scores no hits at $n$=40, and this paper's own argument is that a failed attack bounds what that attack could express rather than certifying absence. Protecting all PII fields jointly is therefore far more expensive than protecting one, and the single-field operating point is the one that stays deployable.

### E. Generalization to unseen identities

A natural objection is circularity: if $D$ is estimated from the same values the attacker later queries, the defense is scoring against itself. It is not. $D$ is fit on a 180-name training split while every victim record is drawn from a disjoint held-out split, so the defense is evaluated open-set by construction.

We tested the stronger condition as well. Fitting $D$ on one 240-name vocabulary and drawing *both* the victims and the entire attack lineup from a second vocabulary sharing no first name and no surname with the first, $k$=96 still yields adaptive recovery 0.100 at utility 0.875, identical to the in-vocabulary

TABLE IX

CATARACT ($k$=96) ON THE SPANISH (ESP) SUBSET OF REALISTIC IDENTITY DOCUMENTS, FIT ON THE TARGET CORPUS ($n$=60, $K$=200, CHANCE 0.005). THE UNDEFENDED COLUMN IS THE ESP ROW OF TABLE III. IDENTITY-NUMBER RECOVERY SPANS 0.217–0.933 ACROSS THE FIVE LOCATIONS IN THAT TABLE, SO THIS IS ONE LOCATION RATHER THAN AN AVERAGE OVER THEM.

| Field | Undefended | CATARACT | Suppression |
|---|---|---|---|
| name | 0.750 | **0.017** | −0.733 |
| ID no. | 0.900 | **0.017** | −0.883 |
| DOB | 0.867 | **0.000** | −0.867 |

sweep, with undefended recovery still 1.000. CATARACT generalizes to identities sharing no tokens with its fitting set.

## X. EVALUATION

### A. Does it work on real documents?

Until now the defense's privacy had been measured only on synthetic cards, and the single real-document data point was negative: a CATARACT fit on synthetic cards and applied to FUNSD suppressed linkage only from 0.221 to 0.169 (the defense-transfer subset, 272 fields over 40 pages, against the 449-field answer set of Sec. IV). Table IX resolves that by fitting $D$ on the target corpus's own values and re-running the attack through it on realistic identity documents.

Linkage falls to near the 0.005 chance floor on realistic imagery, stronger than the 0.10 obtained on synthetic cards. Two further observations follow. $D$ is estimated from *name* values alone, yet identity numbers and dates are suppressed equally hard, independent support for the holographic mechanism, and consistent with the shared-subspace result of Sec. IX-D, since a per-field subspace would not transfer this way. And read together with the FUNSD result, the earlier failure was **distribution shift, not defense failure**: the deployment rule is to fit CATARACT on the corpus being protected.

### B. Retrieval utility on a real corpus

Synthetic topic Recall@1 is not a retrieval metric. On a real ViDoRe corpus [1], CATARACT at $k$=96 takes NDCG@5 from 0.630 to 0.542 and Recall@1 from 0.533 to 0.433, 86.2% NDCG@5 retention. Matched flat Gaussian noise reaching comparable utility ($\sigma$=0.1 → 0.547) buys far less privacy. Real ViDoRe carries no PII labels, so this measures utility cost only; privacy on real documents is Table IX.

### C. Comparison to prior defenses

Table X places CATARACT against prior embedding-privacy defenses and against dimensionality reduction.

The dimensionality baselines answer the obvious alternative, why not simply store smaller embeddings? Compressing each patch to 32 dimensions by random projection or PCA still leaks at 0.175 and 0.20. Compression is content-agnostic; it discards capacity uniformly rather than removing the directions that carry identity. **The win is subspace removal, not fewer dimensions.** Only PRESS, itself a subspace method, buys comparable privacy, and at greater utility cost and only non-adaptively.

TABLE X

DEFENSES ON THE SAME FRONTIER. **THESE ARE OUR RE-IMPLEMENTATIONS, NOT PUBLISHED NUMBERS.** WE PORT EACH MECHANISM TO AN INDEX-TIME PATCH TRANSFORM, SWEEP ITS STRENGTH KNOB OVER OUR HELD-OUT CORPUS, AND SCORE IT WITH THE METRIC OF SEC. II-C; EACH WAS PROPOSED FOR A SINGLE-VECTOR SETTING AND REPORTS FIGURES THAT ARE NOT COMPARABLE TO THESE. THE COLUMNS ARE ALSO NOT LIKE-FOR-LIKE, AND THAT ASYMMETRY IS THE POINT: CATARACT IS MEASURED AGAINST AN ADAPTIVE ATTACKER, THE OTHERS AGAINST THE NON-ADAPTIVE ATTACK EACH WAS DESIGNED FOR, THE MODEL IN WHICH SEC. VIII SHOWS A DEFENSE CAN SCORE 1.000 PRIVACY WHILE PROVIDING NONE. WE DO NOT CLAIM TO HAVE BROKEN ANY OF THEM IN THEIR OWN SETTINGS. CATARACT IS THE ACROSS-SEED MEAN OF TABLE VII.

| Defense | Privacy | Utility | Attacker |
|---|---|---|---|
| EntroGuard [9] | 0.00 | 0.93 | non-adaptive |
| Koga et al. [10] | 0.03 | 0.93 | non-adaptive |
| PRESS [11] | 0.82 | 0.80 | non-adaptive |
| Random proj. (32-d) | 0.175 | 0.85 | non-adaptive |
| PCA (32-d) | 0.20 | 0.925 | non-adaptive |
| **CATARACT ($k$=96)** | **0.892** | **0.875** | **adaptive** |

### D. Deployment cost

CATARACT is a fixed matrix multiplication applied once per page at index time: 0.64 ms per page, +0 bytes of storage, and zero query-path overhead. Protecting $10^9$ pages costs roughly 178 CPU-hours as a one-time operation. The only recurring cost is the tunable utility trade.

### E. The attacker needs the index encoder

The threat model assumes the public encoder that produced the index. That assumption is load-bearing rather than convenient: an attacker substituting a *different* encoder (ColPali index queried with ColQwen2, ridge-aligned on 256 anchors) recovers 0.000 in both directions. The attack is white-box on public weights, not model-agnostic transfer, a meaningful limit on who can mount it, and one we state rather than eliding.

## XI. RELATED WORK

*Privacy in retrieval systems:* Work on retrieval privacy has so far attacked the pipeline's *ends*. Kulkarni et al. [12] mount membership inference against retrieval-based in-context learning for document question answering, inferring whether a document is in the store; Xiang et al. [13] defend the same pipeline against corrupted retrievals, the integrity counterpart to our confidentiality question. Closest in spirit, Xin et al. [14] show that sanitized text leaks semantically: surface PII removal survives re-identification because auxiliary detail still links records. Their released asset is text, and embeddings appear only as the attacker's retriever; ours is the reverse (the embeddings are the leaking artifact, never rendered back into text or image) and where they close by calling for better methods, we supply and certify one. BinaryShield [15] illustrates what is at stake: a deployed pipeline that redacts PII, embeds, and quantizes, on the premise that the resulting

fingerprints are hard to invert. The systematization of RAG privacy [4] names *vector database leakage* as a risk class and finds few mitigations proposed at the embedding level; the systematization of data minimization [6] classes embeddings as a minimization *mechanism*, a transformative encoding that reduces exposure. Our result cuts against that premise for multi-vector storage, and its points of minimization are stages of training and inference, with no index-time point of the kind we add.

*Text-embedding inversion:* Vec2Text [16] showed that dense text embeddings regenerate their source text, recovering $92\%$ of 32-token inputs (independently reproduced in [17]), and GEIA [18] extended generative inversion to sentence encoders. Both target *single-vector text* encoders and free-form language. Our setting differs in the object attacked, a many-vectors-per-page visual index, and in the attack form: we report closed-set linkage *and* open-world composition (Sec. IV-D), and neither requires any training at all.

*Vision-encoder inversion and descriptor privacy:* Leaky-CLIP [19] extracts training data from CLIP, and Implicit Inversion [20] turns CLIP into a decoder. On the defensive side, NinjaDesc [21] learns content-concealing visual descriptors adversarially, and TrustCLIP [22] optimizes features directly against a generative reconstruction attacker. Oz et al. [23] come nearest in method, reconstructing training data in the embedding space of frozen DINO-ViT and CLIP features rather than in pixels; they still target a reconstruction, whereas we show the discriminative probe recovers field values with no reconstruction at all. These operate on pooled image descriptors; none addresses per-patch multi-vector document indices, and none evaluates a defense against an *adaptive* inverse learner, which is where our reshaping defense fails.

*Storage-layer leakage and erasure:* Closest to our threat model, Chakraborttii et al. [24] show that soft-deleted embeddings persist as recoverable "ghost vectors" in production vector-store index files, recovering single-vector text and image embeddings from bytes an operator believed erased, and propose encryption with epoch key rotation. We reproduce a multi-vector analogue (logical deletion returns $0.000$ while the raw segment still yields $1.000$) but we claim only the multi-vector extension. The storage-layer erasure gap and its Article 17 significance were established there. Our measurement uses an in-process index reproducing tombstone-without-erase semantics, *not* a production database at a pinned version, and we do not claim byte-identical recovery from a named system. Certified removal [25], [26] supplies the formal vocabulary for what deletion ought to guarantee, and exact-unlearning constructions motivated by the same right [27] show how much machinery it takes to honour it inside a model. Our point is that the index is a second place the obligation lands, and that at the storage layer the usual remedy (deleting what overlaps the field) does not work at all.

*Concept erasure and linear subspace removal:* The transform in Sec. IX belongs to an established family, and we make the inheritance explicit. INLP [28] removes a concept from a representation by repeatedly fitting a linear probe and projecting onto its nullspace; R-LACE [29] recasts that objective as a constrained minimax game over rank-$k$ projections; LEACE [30] gives a closed-form projection that provably defeats *every* linear probe while perturbing the representation minimally under a broad class of norms. CATARACT is INLP's construction moved to index time: we likewise estimate a discriminative subspace and project it away, and the irreversibility argument of Sec. IX is the ordinary one for an orthogonal projection rather than a new guarantee. What differs is the setting and what it demands. Those methods erase a low-cardinality attribute, gender or part of speech, from one vector per example. We erase open-vocabulary field values from a $1024$-vector-per-page visual index, where the quantity to remove is not a label but the identity-discriminative directions shared across name, identity number and date (Sec. IX-D), where removal must hold against an attacker who knows the projection, and where the operator pays for it in retrieval quality rather than probe accuracy. LEACE's guarantee is strictly stronger than ours against linear adversaries, and an operator who can afford its whitening step should prefer it; we report the SVD estimator because it is the one our certification and our adaptive evaluation are computed for. None of this literature has been applied to retrieval indices, and none of it is evaluated against an adaptive inverse learner.

*Embedding-privacy defenses:* EntroGuard [9] perturbs embeddings via entropy-driven noise; Koga et al. [10] apply differential privacy to retrieval-augmented generation; PRESS [11] shifts embeddings in a learned subspace. Table X evaluates all three on our frontier. The decisive difference is not mechanism but *evaluation*: each is reported against a non-adaptive attacker, and Sec. VIII shows that a defense can score $1.000$ privacy in that model while providing none at all. This is the failure mode adversarial robustness took years to internalise: defenses that survive the attacks they were tested against and fall to ones built for them [31], [32], with adaptive evaluations that are incomplete exactly where they are most confident [33]. The same critique has since been levelled at model fingerprinting, where schemes evaluated only against benign perturbations fall to attacks tailored to each scheme's specific weakness [34]. Our inverse learner is that attack for this class, and we report CATARACT against it rather than against the attack CATARACT was built for.

*Token pruning:* Efficiency work reduces the number of retained tokens for speed, ToMe [35], EViT [36], DynamicViT [37]. Sec. VI shows that reducing retained capacity also reduces linkage, so pruning is an incidental privacy lever. It is a poor one: it discards capacity content-agnostically, which is exactly why the dimensionality baselines in Table X still leak.

## XII. DISCUSSION, SCOPE, AND ETHICS

*What the results constrain:* Retrieval quality and field-level exposure are entangled in this architecture. Linkage tracks retained per-patch capacity, and that capacity is the property delivering state-of-the-art retrieval. No defense removes the leak while keeping what motivates the design; it can only trade along the frontier, and CATARACT's cost (three quarters of the

embedding dimensions) is the price of that trade rather than an artifact of our construction.

*Scope and open directions:* Each item below states what the present results cover and the work it opens.

*From lineups to vocabularies.* Most numbers here are closed-set and are read against their stated $K$. Removing the lineup entirely still composes the correct name on $0.147$ of documents from public vocabularies, and the remaining step (a scorer that ranks whole names jointly rather than scoring parts independently) is the direction that would raise it.

*From linear to nonlinear certification.* Eq. 5 covers linear reconstruction. A trained nonlinear inverse recovers roughly $0.73$ of the removed span component; end-to-end linkage still collapses, and extending the certificate to arbitrary nonlinear adversaries is the natural next theorem.

*Toward chance-level residual linkage.* At $k{=}96$ residual linkage is $0.125$ on the real vocabulary at $n{=}200$, an order of magnitude below the undefended rate and roughly $25\times$ the $K{=}200$ chance floor. Closing the remaining gap calls for an encoder trained so that identity occupies few removable directions, rather than a larger projection at index time.

*Fitting the subspace on the corpus being protected.* Fit on its target distribution, CATARACT reaches $0.017$ linkage on realistic identity documents, below the operating point we headline. Fit instead on synthetic cards and applied to FUNSD it suppresses only from $0.221$ to $0.169$. The subspace is therefore a property of the corpus rather than of the architecture, which gives operators a clear recipe: fit on the index you are protecting. Doing so needs unlabelled pages from that corpus and one pass of the encoder.

*Where the frontier is observed jointly, and where it is not.* The frontier itself is a single-corpus measurement: every row of Table VI scores privacy and utility on the *same* synthetic cards, the same documents and the same seeds, so the trade at each $k$ is observed rather than inferred, and the operating point of Table VII pairs $0.108$ linkage with $0.875$ utility on one corpus. What is measured separately is the *real*-corpus evidence on each side, because no public corpus we are aware of carries both kinds of label: IDNet and our cards carry field labels but no relevance judgements, ViDoRe carries relevance judgements but no PII. Real-document privacy (Table IX) and real-corpus utility ($86.2\%$ NDCG@5 retention) are therefore established on different corpora, and we report them as such. A corpus carrying field labels and relevance judgements together would let the real frontier be traced directly, and would serve every defense in this class, not only ours.

*Retention at deployment scale.* Recall@1 is harder to retain as the evaluation set grows: CATARACT retains $88\%$ over $40$ documents and $78\%$ over $200$, while NDCG@5 on the real ViDoRe corpus holds at $86.2\%$. We report the trend rather than extrapolate it, and the released harness measures retention at any corpus size, so an operator can establish the figure for their own index before deploying.

*Selecting $k$ without labels.* The operating point $k{=}96$ is selected on the sweep it is reported on, and is stable across every seed. A label-free criterion for choosing $k$ (a spectral test on $D$, say) would let an operator tune it on a corpus for which no PII probes exist.

*Scope of the corpora.* IDNet documents are realistically rendered but synthetically populated, so no real person's data is involved at any point. Our five national templates are European; extending the measurement to non-Latin scripts is open, and the glyph-height law of Sec. IV predicts how it should behave. Identity-number recovery spans more than fourfold across the five templates ($0.217$–$0.933$) and should be read as a range, not a point.

*Storage-layer scope.* The soft-delete result is measured on an in-process index model, not a production vector database. Production stores do support native multi-vector $\mathrm{MaxSim}$ collections [38], so the setting is realistic, and measuring a named production store at a pinned version is a direct extension of the released harness.

*Ethics and disclosure:* All experiments used synthetic identity cards, the public FUNSD form fields, and IDNet's synthetically populated documents. No real person's data was collected, stored, or inverted at any point, and IDNet's realistic rendering carries generated field values rather than genuine personal information. Our intent is defensive: we pair the vulnerability with a countermeasure that costs $0.64\,\mathrm{ms}$ per page and no storage.

We notified the maintainers of the affected retrievers, and of a widely deployed vector database offering multi-vector collections, on 5 September 2026, describing the attack, its scope, and the mitigation, and offering the draft in advance. We were explicit in both notifications that the finding is an architectural property rather than a defect in either codebase, and that our storage-layer measurement is made against an in-process index model rather than a production deployment.

We assess the residual risk of releasing this work as low, and say why. The attack requires read access to a stored index, which an adversary must already have obtained by other means; the technique is a direct application of the retriever's own scoring function rather than a novel capability; and the mitigation is published alongside it, is cheap, and requires no change to the query path. The artifact reaches reviewers through the anonymized repository described in the Open Science section, and will be released publicly alongside the paper.

*Future work:* Two directions follow directly. *Open-world extraction*, beam-search decoding over the token space, would remove the closed-set restriction and is the natural strengthening of the attack. *Privacy-native retrieval*, training the encoder so that identity concentrates into a few cheaply removable dimensions, would attack the entanglement itself rather than paying its price, and is the only route we see to privacy without the utility trade.

## XIII. CONCLUSION

Stored multi-vector embeddings are not an opaque summary of a document; they are a field-level oracle for the personal data the document contains. This paper establishes four things about that oracle.

First, it can be read. A training-free dictionary attack over the at-rest index and the public encoder recovers names, identity numbers and dates at $0.994/0.804/0.794$ top-1 from a 200-candidate lineup drawn from a real $6.8\times10^8$-name vocabulary, and composes the correct full name on $0.147$ of documents with no lineup at all. Second, the leak is holographic: deleting a field's own patches, or a third of the page around them, leaves recovery at $1.00$, so patch-level redaction and erasure do not reach the index. Third, it is entangled with the property the architecture is deployed for, linkage rises monotonically with retained per-page capacity, so it cannot be engineered away while keeping what makes these retrievers state of the art. Fourth, a defense must destroy rather than reshape. A learned residual transform reports $1.00$ privacy against a non-adaptive attacker and returns exactly the undefended rate against an adaptive one. CATARACT projects stored patches off the PII-discriminative subspace instead, and holds that same attacker to $0.125$ linkage while retaining $86\%$ of NDCG@5 on a real retrieval corpus, for $0.64$ ms per page and no added storage.

The consequence for practice is immediate: an index built from visually rich documents contains the personal data of those documents, and should be governed, audited and erased as such. That governance is affordable today. One index-time projection, costing $0.64$ ms per page, no query-path work and no extra storage, moves field-level exposure by an order of magnitude, and an operator can fit it on their own corpus with unlabelled pages and a single encoder pass. The larger prize is an encoder trained so that identity occupies few removable directions, retiring the utility trade rather than paying it; the measurements and the released harness here are what make that target concrete and testable.

## OPEN SCIENCE

The artifact contains the attack and CATARACT implementations, the synthetic identity-card generator, the driver for each experiment, the aggregation scripts that build every table, the figure sources, and the recorded output of every run. Results are keyed to the git revision and container image that produced them, so a table entry can be traced to the code path behind it.

Reproduction has two tiers. The GPU tier re-runs the encoders and regenerates every result from images on one 40 GB-class accelerator. The CPU tier needs no accelerator: the recorded outputs ship with the artifact, so every table and figure is re-derivable on a laptop in about a minute, and a suite of $247$ tests covering the attack, the CATARACT construction, the certified bound and the statistics layer runs in seconds without loading a retriever.

No access-restricted data is involved. FUNSD and IDNet are public; the identity cards are generated by the released script from a seeded synthetic vocabulary, so no component contains or reconstructs a real person's data. The artifact is provided at submission through an anonymized repository and will be deposited on Zenodo with a DOI at camera-ready.

## LLM USAGE CONSIDERATIONS

An LLM was used under the authors' direction for editorial assistance on the manuscript, for code review and occasional code edits, and for running experiments the authors designed and aggregating their outputs. The research question, the threat model, the system architecture, the experimental design, the implementation and the manuscript are the authors' own. The authors are responsible for every result, line of code and citation in this paper.

LLMs were used for editorial purposes in this manuscript, and all outputs were inspected by the authors to ensure accuracy and originality.

No number in this paper rests on an LLM's report of it. Every result is produced by executed code whose raw outputs ship with the artifact, each tagged with the git revision and container image that produced it, so any table entry can be re-derived independently of this text. Every citation was checked against its source: arXiv identifiers resolved directly, and author lists for recent venue papers taken from the venue's published programme.

All experiments ran on a single NVIDIA A100-SXM4-40GB or a single NVIDIA L4. The retrievers we study are three-billion-parameter models run in bfloat16, which fits either device; no LLM was trained. Query cost is dominated by encoding candidate strings, so each candidate pool is encoded once, cached, batched, and reused across every lineup size and document.

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## APPENDIX A
## THE CATARACT TRANSFORM

Algorithm 1 states the index-time transform in full. It is applied once, to the bytes actually written, and adds no parameters to the query path and no bytes to storage. Steps 1–4 are executed once per corpus; step 6 is the per-patch operation applied at write time.

**Algorithm 1** CATARACT index-time redaction

**Require:** encoder $E$; fitting names $\mathcal{N}$; topic queries $\mathcal{T}$; rank $k$; spared content rank $r$
**Ensure:** projection $\Pi$; redacted patches for every stored page
1: $N \leftarrow [\, E(v) \;:\; v \in \mathcal{N} \,]$ ▷ name-value query encodings
2: $T \leftarrow [\, E(t) \;:\; t \in \mathcal{T} \,]$ ▷ topic query encodings
3: $V_r \leftarrow \textsc{TopDirs}(T, r)$ ▷ content directions to spare
4: $D \leftarrow \textsc{TopDirs}\big(N - NV_rV_r^\top,\, k\big)$ ▷ PII-discriminative basis
5: $\Pi \leftarrow I - DD^\top$
6: **for all** stored patch $x$ of every page **do**
7: $\quad x' \leftarrow \Pi x / \|\Pi x\|$
8: $\quad$ **write** $x'$ in place of $x$
9: **end for**
10: **return** $\Pi$

$\textsc{TopDirs}(M, r)$ returns the top-$r$ right singular directions of $M$ as an orthonormal $d \times r$ matrix. Because $\Pi$ is a rank-deficient orthogonal projection, step 7 is many-to-one on $\mathrm{span}(D)$: the removed component is absent from the stored bytes rather than concealed in them.

## APPENDIX B
## COMPLETE CATARACT SWEEP AND PER-FIELD SUBSPACES

Table VI reports the operating points that matter; the full sweep including the degenerate endpoint is given here for completeness. At $k{=}128$ every dimension is removed: non-adaptive privacy reads $1.000$ but utility is $0.000$, and the inverse-attack figure at that point is not meaningful because the representation is degenerate. We exclude it from the frontier rather than reporting a privacy number obtained at zero utility.

## APPENDIX C
## CONTROLS IN FULL

Sec. IV-B summarizes the controls; the complete results are here.

*Wrong-page:* Synthetic: correct-page recovery $1.000$, wrong-page $0.000$ (exact, $n{=}50$, chance $0.005$), both with the field's own patches deleted and intact. Realistic imagery: wrong-page $0.000$ / $0.017$ / $0.000$ for name / ID / DOB against attack recoveries of $0.750$ / $0.900$ / $0.867$.

*Anagram:* The true value must beat character permutations of itself, or $\mathrm{MaxSim}$ is bag-of-tokens matching. Names $1.00$; dates $0.93$–$1.00$; identity numbers $0.77$–$0.87$, with residual errors being digit transpositions.

*Arrangement null:* Shuffling field order while holding content fixed leaves recovery at $1.00$, so the attack keys on content rather than spatial arrangement.

*Cross-encoder transfer:* An attacker substituting a different encoder (ColPali index queried with ColQwen2, ridge-aligned on 256 anchors) recovers 0.000 in both directions.

*Token-disjoint vocabulary:* $D$ fit on one 240-name vocabulary, with victims and the entire lineup drawn from a second sharing no first name and no surname: $k$=96 gives adaptive recovery 0.100 at utility 0.875, matching the in-vocabulary sweep, with undefended recovery 1.000.

*A control we discarded:* An earlier attempt to remove lineup overlap padded the candidate set with procedurally generated names to hold $K$ fixed. It reported $k$=96 recovery of 0.450. We do not report that number as a result: procedurally generated names are distributionally distinguishable from real ones, so the attacker can win on "which candidate looks like a real name" rather than on identity. The token-disjoint vocabulary control above replaces it. We record the failure because the lesson generalizes: **any lineup control must keep distractors in-distribution.**

## APPENDIX D
## REPRODUCIBILITY

Every experiment runs inside a container image pinned by git SHA, and each result file records that SHA together with torch and CUDA versions and the accelerator used (NVIDIA A100-SXM4-40GB or NVIDIA L4), so any number in this paper can be traced to the code that produced it. Seeds are fixed and recorded; headline results are repeated across three seeds and reported with bootstrap confidence intervals and Holm–Bonferroni correction.

Two aggregation hazards are worth recording, since both silently produce wrong numbers rather than errors. First, the seed-0 sweep contains one $k$ value the later sweeps do not, so aggregating by list position averages $k$=80 with $k$=96 and reports 0.175 where the correct answer is 0.108; aggregate by $k$ value, for which the released aggregator takes a `sweep[k=96]` selector. Second, monotonicity must be judged on unsaturated fields, *name* sits at the ceiling on every multi-vector model, so a single flipped document reverses the verdict while the informative fields rise cleanly.

The artifact itself, and what can be reproduced with and without a GPU, are described in the Open Science section.